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Institute of Bioinformatics and Applied Biotechnology (IBAB)

Term-End Examination - M.Sc. BDB Batch 01 - Second Semester May 2024

BDBP207: Machine Learning Laboratory

Time: 4 hours **Maximum Marks: 70**

General Instructions

1. **Please do not use Google, AI tools or the class materials for this test.**
2. **Please send your answers as a single .py file to shyam@ibab.ac.in**
3. **Marks will be given to correctness, handling all the test scenarios, code formatting (indents, variable names), user-friendly messages.**

PART - A

Answer any 1 question. It carries 10 marks

1. Consider the USArrests data. Perform hierarchical clustering on the states.

   (a) Using hierarchical clustering with complete linkage and Euclidean distance, cluster the states.

   (b) Cut the dendrogram at a height that results in three distinct clusters. Which states belong to which clusters?

   (c) Using hierarchical clustering with complete linkage and Correlation based distance, cluster the states.

   (d) Cut the dendrogram at a height that results in three distinct clusters. Which states belong to which clusters?

2. Let $x = [x_1, x_2, x_3]$, and $y = [y_1, y_2, y_3]$. Implement a feature mapping function, $\phi(x)$, called Transform(x), given by $\phi(x) = (x_1x_1, x_1x_2, x_1x_3, x_2x_1, x_2x_2, x_2x_3, x_3x_1, x_3x_2, x_3x_3)$.

   a. Let x = [1, 2, 3], y = [4, 5, 6]. Use the above "Transform" function to transform these vectors to a higher dimension and compute the dot product in a higher dimension. Print the value.
   b. Implement a kernel, $K(x,y) = (< x,y >)^2$. Apply this kernel function and evaluate the output for the same x and y vectors. Show that the result is the same in both scenarios demonstrating the power of a kernel trick.

PART B

Answer any 4 out of the following 5 questions. Each question carries 15 marks.

1. In this problem, you will create some simulated data and will fit a simple linear regression model to it. Make sure to use the default random number generator with seed set to 1 prior to starting part (a) to ensure consistent results.

(a) Using the normal() method of your random number generator, create a vector, X, containing 200 observations drawn from a N (0, 1) distribution. This represents a feature, X

(b) Using the normal() method, create a vector, e (noise), containing 200 observations drawn from a N (0, 0.25) distribution—a normal distribution with mean zero and variance 0.25.

(c) Using X and e, generate a vector y according to the model $y = -1.1 + 0.6X + e$,

where e = mean-zero random error/noise term. What is the length of the vector y?

What are the values of theta_0 and and theta_1 in this linear model? Answer these

questions as a comment in the code.

(d) Create a scatter plot displaying the relationship between x and y. Comment on what you observe.

(e) Fit a least squares linear model to predict y using X after forming a 70-30 split. Create a scatter plot displaying X_test and y_test. Draw the linear regression line in red color based on the X_test and the obtained y_pred.

2. For the heart.csv dataset, build a logistic regression classifier to predict the risk of heart disease. Vary the threshold to generate multiple confusion matrices. Implement a python code to calculate the following metrics

- Accuracy
- Precision
- Sensitivity
- Specificity
- F1-score
- Plot the ROC curve
- AUC

3. This question should be answered using the Weekly data set, which is part of the ISLP package. Implement the Leave-One-Out-Cross-Validation (LOOCV) algorithm.

(a) Write a for loop from i = 1 to i = n, where n is the number of observations in the data set, that performs the following step: Fit a logistic regression model using all but the ith observation to predict Direction using Lag1 and Lag2. The 'i'th observation is the test data and the remaining 'n-1' observations are the training data.

(b) Take the average of the n numbers obtained in (a) in order to obtain the LOOCV estimate for the test accuracy. Print the estimated average test accuracy.

4. This problem involves the OJ data set which is part of the ISLP package.

(a) Create a training set containing a random sample of 1000 observations, and a test set containing the remaining observations.

(b) Fit a support vector classifier (svm.LinearSVC) to the training data using C = 0.01, with Purchase as the target and the other variables as features. Measure the train and test accuracies.

(c) Use GridSearchCV to select an optimal C. Consider values in the range 0.01 to 10.

(d) Compute the training and test error rates using this new value for C.

(e) Fit a support vector machine with a radial kernel. Use the default value for gamma and measure the train and test accuracies.

Overall, which approach seems to give the best results on this data?

5. In this problem, you will generate simulated data, and then perform PCA and K-means clustering on the data.

(a) Generate a simulated data set with 40 observations in each of three classes (i.e. 120 observations total), and 50 variables. Hint: There are a number of functions in Python that you can use to generate data. One example is the normal() method of the random() function in numpy; the uniform() method is another option. Be sure to add a mean shift to the observations in each class so that there are three distinct classes.

(b) Perform PCA on the 120 observations and plot the first two principal component score vectors. Use a different color to indicate the observations in each of the three classes. If the three classes appear separated in this plot, then continue on to part (c). If not, then return to part (a) and modify the simulation so that there is greater separation between the three classes. Do not continue to part (c) until the three classes show at least some separation in the first two principal component score vectors.

(c) Perform K-means clustering of the observations with K = 3. How well do the clusters that you obtained in K-means clustering compare to the true class labels? Hint: You can use the pd.crosstab() function in Python to compare the true class labels to the class labels obtained by clustering. Be careful how you interpret the results: K-means clustering will arbitrarily number the clusters, so you cannot simply check whether the true class labels and clustering labels are the same.